

ALPACA AI TRADING AGENTS HACKATHON

Vetoed

An autonomous options agent where **the AI is the least-trusted component.**

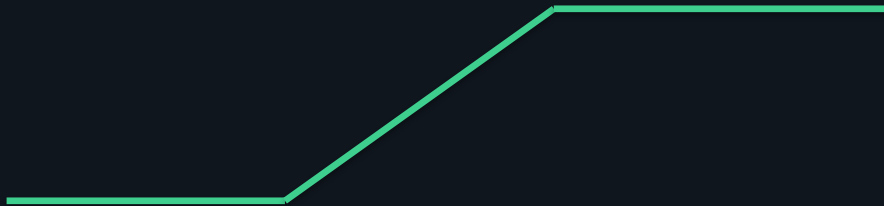
Alpaca paper trading · [chong1120.github.io/Vetoed](https://github.com/chong1120/Vetoed)

We sell defined-risk insurance.

Two legs, always. The one we sell collects a premium; the one we buy **caps the loss before the order is sent.**

PUT CREDIT SPREAD

Profits if the stock does not fall far



SELL the 310 put · BUY the 305 put

CALL CREDIT SPREAD

Profits if the stock does not rise far



SELL the 300 call · BUY the 305 call

Two numbers. When they disagree, the seller is being paid.

Implied volatility — how much movement option prices are charging for.

Realised volatility — how much the stock actually moved recently.

When implied sits above realised, options are priced for more movement than has been happening. **That gap is what a premium seller is trying to collect.**

AAPL, ONE LIVE SCREEN

implied



realised, 20 day



ratio

1.26

One model. Two volatility inputs.

Same spread, same payoff, same probability model. **Only the volatility changes.**

PRICED AT REALISED VOL

+\$28.59

what it is worth if volatility keeps behaving like it has

PRICED AT IMPLIED VOL

-\$13.85

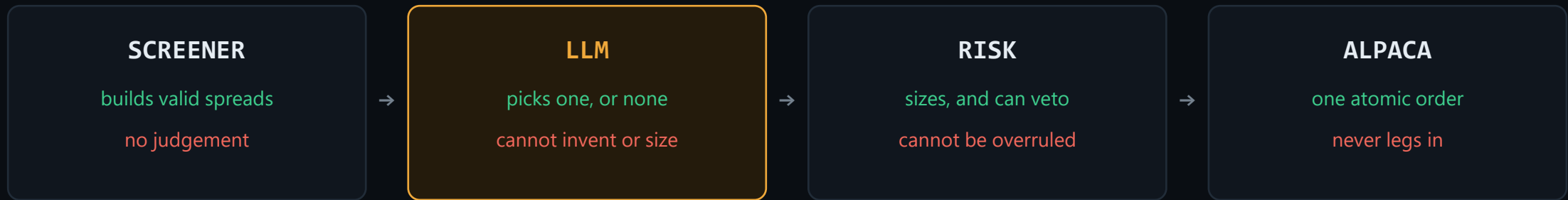
roughly what the market is charging for it today

VETOED'S MODEL-DERIVED EDGE — THE DIFFERENCE

+\$42.45 per spread

This is **our own operational signal**, motivated by volatility-risk-premium research — not the canonical academic VRP, which is defined on variance swap rates over a matched horizon.

The model proposes. Deterministic code disposes.



Echoed contracts are checked against the shortlist

HALLUCINATION → NO TRADE

The model's requested size is discarded

CANNOT SIZE

No broker or tool output ever enters the prompt

CONTROLLED VOCABULARY

No model available? Arithmetic selection runs instead

STILL AUTONOMOUS

RISK IS NOT ADVISORY

Eight gates. Any one of them rejects the trade outright.

5%

of equity at risk
in any one spread

-3%

on the day halts
the whole session

25

contracts, hard cap
whatever the maths says

0

naked positions,
structurally impossible

2

legs, one atomic order —
both fill or neither

236

tests, most of them
on the gates

WHAT I AM NOT CLAIMING

The limits, stated before you find them.

- **Quotes are indicative**, not true NBBO.
- **Four correlated tickers** is not diversification.
- **Skew isn't modelled** — both legs use one volatility.
- **Exit thresholds are unvalidated.**
- **20-day realised vol is an estimator**, not a forecast of volatility to expiry.
- **A contest week proves nothing** about profitability. The literature motivates the idea; it does not validate this system.

An agent that's most useful when it says no.